

分段刷新延迟需求回复 / Reply: Segment Refresh Delay Request

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 - 收件 / To: R2 Engineering (Hirata)
 - 发件 / From: LIN
 - 参考 / Reference: 《显示控制协议_16色.md》V1.4 §7.4 / §7.4.7 (英文版 / English version: Display Control Protocol_16-Color_EN.md)
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Q1. 0x1E 是否空闲? 若否, 应使用哪个命令码? / Is 0x1E free? If not, which code should we use?

不空闲。 0x1E 已被"切换下一槽位"命令占用 (主从多机同步切槽的核心命令, 不可挪用)。请勿向 0x1E 发探测帧——在从机模式下会误触发播放并返回 ACK_SUCCESS (有副作用)。

No. 0x1E is already occupied by the "switch to next slot" command (a core command for master/slave slot synchronization; it cannot be repurposed). Do not send probe frames to 0x1E — in slave mode this would mistakenly trigger playback and return ACK_SUCCESS (side effects).

请使用 0x1F (下发逐段 Pipeline): 按槽位下发"每段刷新起始延迟表", 与颜色数据独立、按槽位配对存储; 配套 0x25 可单独清除某槽的表 (不动颜色数据)。格式与示例见协议 V1.4 §7.4。

Please use 0x1F (Send Per-Segment Pipeline): it sends a per-segment refresh start-delay table for a given slot, stored independently of the color data and paired with it per slot; the companion command 0x25 clears a slot's table alone (color data untouched). For format and examples, see protocol V1.4 §7.4.

Q2. 两个"同时"启动的分段之间的最小间隔 (功耗限制) 是多少? / Minimum spacing between two segments that start "at the same time" (power limit)?

无强制最小间隔。 延迟值相同的段在同一帧启动 (允许真正同时刷新); 错峰的最细粒度为 **1 帧 (标称 10 ms, 精确时距以实测为准)**。

No enforced minimum spacing. Segments with the same delay value start in the same frame (truly simultaneous refresh is allowed); the finest staggering granularity is **1 frame (nominally 10 ms; exact timing subject to physical measurement)**.

Q3. 100 ms 单位、最大 25.4 s 是否合适? 50 ms 单位 (最大 12.7 s) 是否也容易实现? / Is a delay unit of 100 ms and a maximum of 25.4 s acceptable, or would 50 ms units (max 12.7 s) be easy as well?

已按更优方案实现: uint16、单位 = 1 帧 (标称 10 ms), 范围 0 ~ 65535 帧 ≈ 0 ~ 655 s, 远超 25.4 s 需求。 100 ms (=10 帧) 与 50 ms (=5 帧) 均可直接表达, 无需特换单位。

An even better scheme has been implemented: uint16 with a unit of 1 frame (nominally 10 ms), range 0-65535 frames ≈ 0-655 s — far beyond the 25.4 s requirement. Both 100 ms (=10 frames) and 50 ms (=5

frames) can be expressed directly; no unit conversion is needed.

下发格式：每芯片拆低/高半帧两帧（帧封套与 0x13 相同，DataLen 66），拼接规则 $\text{delay}[N] = (\text{高半帧}[N] \ll 8) \mid \text{低半帧}[N]$ ，见协议 §7.4.1。

Transfer format: each chip is sent as two frames — a low and a high half-frame (same frame envelope as 0x13, DataLen 66); the assembly rule is $\text{delay}[N] = (\text{high_half}[N] \ll 8) \mid \text{low_half}[N]$. See protocol §7.4.1.

Q4. 表存 Flash 还是 RAM? / Flash or RAM storage for the table (see 3.2)?

存非易失存储 (Flash)，掉电不丢。 每槽位独立存储区，与该槽颜色数据配对；显示单张 (0x1D) 或播放 (0x16) 该槽时设备自动加载生效，无需额外命令。

Stored in non-volatile storage (Flash); it survives power loss. Each slot has its own independent storage area paired with that slot's color data; when the slot is displayed (0x1D) or played (0x16), the device loads and applies it automatically — no extra command needed.

补充

大延迟会拉长整轮节奏：**一轮 $\approx$ 最大延迟 + 播放间隔**（间隔自刷新场次结束起算），最大延迟暂时不要超过 30 s $\approx 3000 \times 10$ ms。衣服鳞片类型建议使用通用类型 0x06。

主机设置下从机数量（从机数量=总数-1）

Large delays lengthen the round cadence: **one round $\approx$ max delay + playback interval** (the interval counts from the end of the refresh session). For now, please keep the max delay under 30 s (≈ 3000 frames $\times 10$ ms). For garment scales, we recommend using the Generic device type 0x06. On the master, set the slave count (0x1A); slave count = total number of devices – 1.

操作步骤 / Steps:

1. 烧录新固件（0923 版本）。Flash the new firmware (0923 build).
2. 参考提供的延迟数据文件 pipeline_2s_P2 / pipeline_20s_P2，基于业务编写自己的延迟数据表并下发到指定槽位；类型建议使用通用类型 0x06。Referring to the provided delay data files pipeline_2s_P2 / pipeline_20s_P2, write your own delay table based on your business needs and send it to the target slot; the Generic device type 0x06 is recommended.
3. 启动播放。Start playback.